

It's a small world



Learning objective: To identify common micro-organisms.

Key skill: Complete a quiz.

What do you know about micro-organisms? Try this quiz twice to find out!

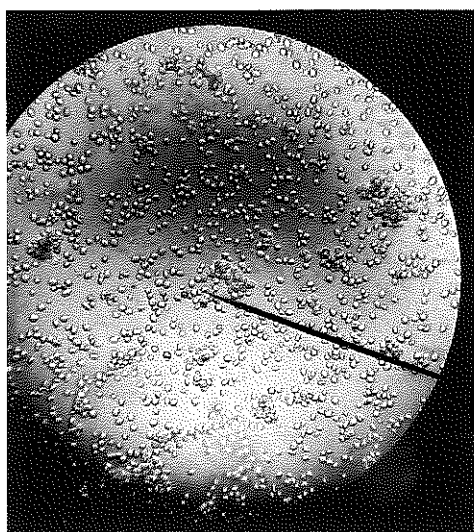
Number correct *before* I learnt about micro-organisms = _____

Number correct *after* I learnt about micro-organisms = _____

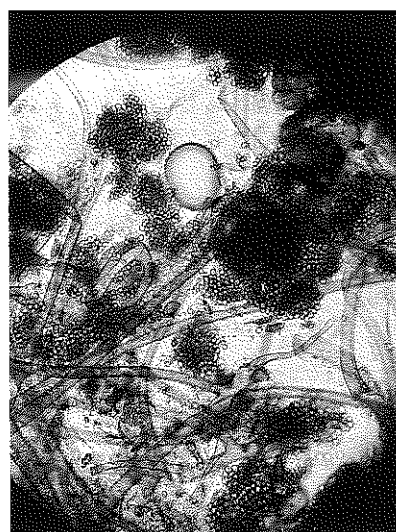
1. What has infected mouldy bread?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
2. What has infected mouldy cheese?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
3. What is the skin infection, athlete's foot?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
4. What is influenza (flu)?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
5. What is tuberculosis (lockjaw or Tb)?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
6. What is chicken pox?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
7. What is bronchitis?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
8. What is a cold that you catch?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
9. What do people use to make bread?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
10. What do people use to make beer?
 - (a) Bacteria
 - (b) A virus
 - (c) Fungi
 - (d) Mould
11. What do vaccinations protect you against?
 - (a) Bacteria
 - (b) Viruses
 - (c) Fungi
 - (d) Mould
12. What is an easy thing you can do to help you avoid catching a virus?
 - (a) Always wear a face mask
 - (b) Use soap and water to wash your hands
 - (c) Use antibacterial soap and water to wash
 - (d) Rest and drink lots of water
13. What is the most effective way to recover from a virus?
 - (a) Get vaccinated
 - (b) Do exercise
 - (c) Rest and drink lots of water
 - (d) Take vitamin supplements

continued ...

14. What does a doctor advise as an effective way to recover from a virus?
(a) Take antibiotics
(b) Rest and drink lots of water
(c) Take vitamin supplements
(d) Get vaccinated
15. What does a doctor advise as an effective way to recover from a bacterial infection?
(a) Take antibiotics
(b) Rest and drink lots of water
(c) Take vitamin supplements
(d) Get vaccinated
16. How can you strengthen your immune system?
(a) Eat healthy food
(b) Exercise regularly
(c) Get vaccinated
(d) All of the above
17. What do you see when you look at a bacterium using an electron microscope?
(a) A cell wall
(b) Circular DNA
(c) A single cell
(d) All of the above
18. What do you see when you look at a virus using an electron microscope?
(a) A cell wall
(b) Circular DNA
(c) A single cell
(d) None of the above
19. What do you see when you look at a yeast cell under a light microscope?
(a) A cell wall
(b) A cell nucleus and DNA
(c) A single cell
(d) All of the above
20. What do you see when you look at a fungus under a light microscope?
(a) A cell wall
(b) A cell nucleus and DNA
(c) Many cells
(d) All of the above
21. What do you see when you look at a mould cell under a light microscope?
(a) A cell wall
(b) A cell nucleus and DNA
(c) Many cells
(d) All of the above
22. How do the micro-organisms we can see under a light microscope compare in size with the micro-organisms we can see under an electron microscope?
(a) They are bigger
(b) They are smaller
(c) They are the same size
(d) They can be bigger or smaller



Baker's yeast under a microscope. Photo: © Sam LaRussa; licensed for reuse under CC BY 2.0



Fungus under a microscope. Photo: © Emilio Bosch; licensed for reuse under CC BY-SA 4.0

Going viral models



Learning objective: To investigate different models that represent a virus.

Key skill: List characteristics of a virus.

Materials: For the paper model: scissors, templates, glue stick, sellotape (single-sided and double-sided); optional – straws, pipe cleaners, string. For the play dough model: play dough, rolling pin; optional – straws, pipe cleaners, string.

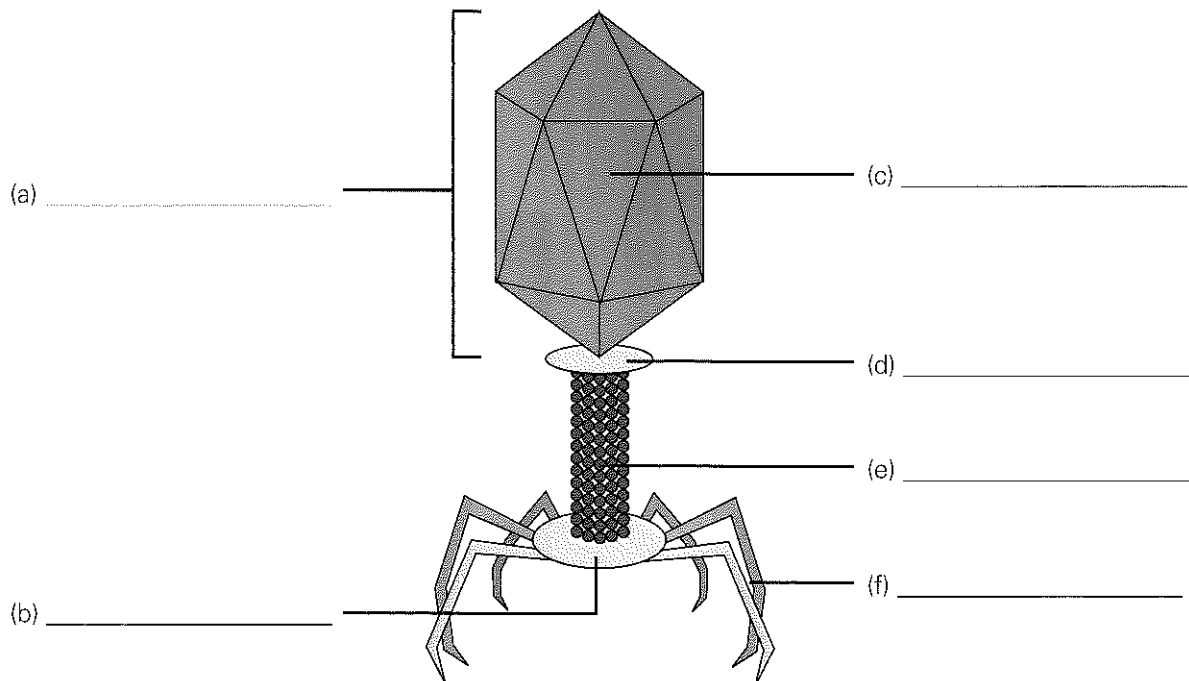
Scientific models make complicated things simple. Here you will explore four models of a virus: a diagram; images; a paper model; and a play dough model.

A virus:

- has RNA or DNA (nucleic acid)
- is a capsid
- may not be living
- is extremely tiny at 20–400 nm
- is not a cell
- has no cell nucleus
- has no cell wall
- may have a tail, collar, sheath, baseplate, fibres and spikes
- may have an external (outside) membrane.

Model 1: Diagram of a virus

Label the diagram, using the word bank to help you.



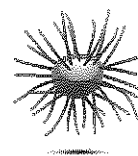
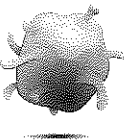
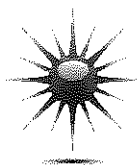
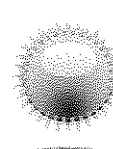
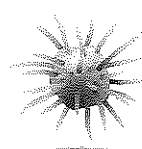
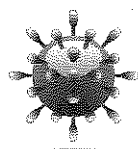
Word bank: capsid, nucleic acid (DNA or RNA), collar, sheath, baseplate, tail fibres

continued ...

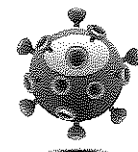
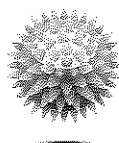
Model 2: Images of a virus

1. All viruses have:

- (a) a capsid True / False
 (b) a collar True / False
 (c) a sheath True / False
 (d) tail fibres True / False
 (e) a collar True / False
 (f) a capsid True / False

2. All viruses have a capsid that contains **nucleic acid**. Nucleic acid can be:

- (a) DNA True / False
 (b) RNA True / False

**Model 3: Paper virus model**

To see paper models of virus and bacteriophage images, go to:

- PDB 101's paper models: <https://pdb101.rcsb.org/learn/paper-models/>
- RCSB Protein Data Banks's viral quasisymmetry: <https://cdn.rcsb.org/pdb101/learn/resources/quasisymmetry/pdb101-all-virus-templates.pdf>
- Wonderstruck's building viruses: <http://wonderstruck.co.uk/resources-and-downloads/Student-Resource-Sheet-2.2-Building-Viruses.pdf>

Model 4: Play dough model of a virus

Make play dough following this recipe and then create your own play dough models of viruses, using what you have learnt from Models 1–3.

Ingredients (for up to four students): 2 tbsp cooking oil, 4 tbsp cream of tartar, 2 cups plain flour, 1 cup salt, food colouring, 2 cups water.**Method:** Mix all the ingredients in a saucepan and stir over medium heat for 3–5 minutes, until the mixture congeals.**Results:** List the viruses you modelled.

How were your playdough viruses similar to the other kinds of models? How were they different?

Going viral quiz



Learning objective: To identify what defines a virus.

Key skill: Complete a quiz.

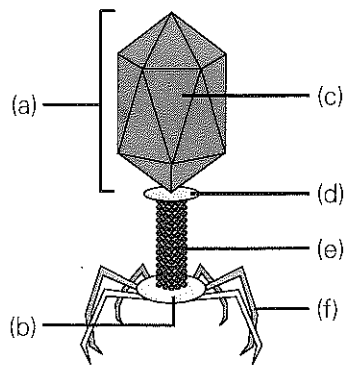
What do you know about viruses? Try this quiz twice to find out! Circle True or False after each statement.

Number correct *before* I learnt about viruses = _____

Number correct *after* I learnt about viruses = _____

- | | |
|--|--------------|
| 1. Viruses are made of cells. | True / False |
| 2. Viruses reproduce on their own. | True / False |
| 3. Viruses contain genetic material. | True / False |
| 4. Viruses can evolve. | True / False |
| 5. Viruses have an outer protein capsid and nucleic acid. | True / False |
| 6. Viruses have an outer carbohydrate capsid and nucleic acid. | True / False |
| 7. Viruses have an outer glyco-protein capsid and nucleic acid. | True / False |
| 8. Viruses may have a protein tail and carbohydrate tail fibres. | True / False |
| 9. Viruses may have a protein capsid, nucleic acid, collar, sheath, baseplate, and tail fibres and spikes. | True / False |
| 10. Viruses may have a membrane envelope with glycoproteins. | True / False |
| 11. Viruses are bacteriophages. | True / False |
| 12. Viruses are extremely tiny. | True / False |
| 13. Viruses eat or engulf bacteria. | True / False |
| 14. Viruses eat or engulf other viruses. | True / False |
| 15. Viruses are much smaller than bacteria. | True / False |
| 16. Viruses reproduce inside host cells. | True / False |

continued ...



17. On the diagram, (d) labels the nucleic acid. True / False
18. On the diagram, (d) labels the capsid. True / False
19. On the diagram, (d) labels the sheath. True / False
20. Viruses are larger than bacteria. True / False
21. Viruses may not be alive. True / False
22. Viruses have a cell wall. True / False
23. Viruses have a cell nucleus. True / False
24. Viruses can contain RNA or DNA. True / False
25. Viruses never contain RNA. True / False
26. Viruses can be killed by antibiotics. True / False
27. On the diagram, (a) labels the capsid, collar and sheath. True / False
28. Viruses may have (f) on the diagram. True / False

Viral infection

How?

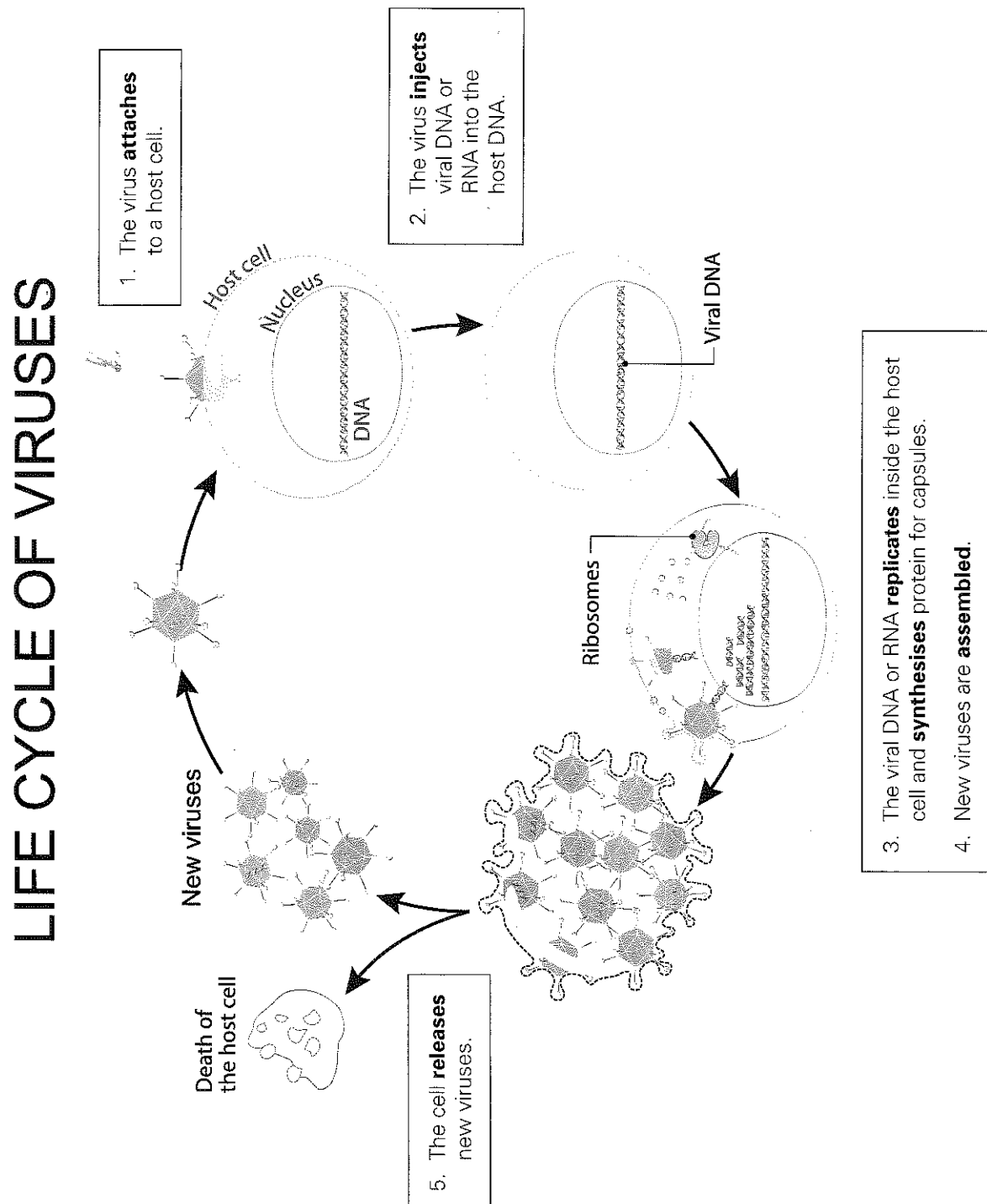
Learning objective: To explore how viruses infect cells.

Key skills: Listen; label a diagram.

Materials: Pencil, eraser. Optional – scissors, glue stick and A4 paper for mix and match.

There are thousands of different kinds of viruses. Viruses cannot be killed by antibiotics.

Bacteriophages are viruses that “eat” bacteria. They are found in the mucus of your mouth, nose and intestine.



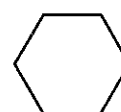
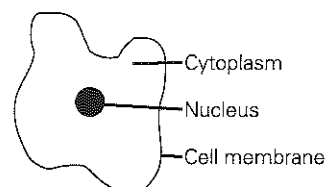
continued ...

Viral infection picture dictation

Your teacher will dictate (read out) a stage of viral infection and a description of the stage twice. Listen the first time, with your pencil on your desk. The second time, draw a diagram in pencil of each of the stages in the table.

Stage	Diagram
Attach	
Inject	
Replicate and synthesise	
Assemble	
Release	

Hint: Use these shapes or similar in your diagrams.



Virus capsule



DNA or RNA

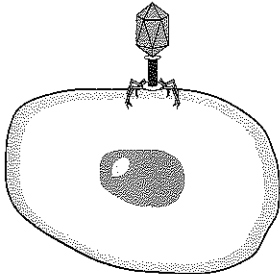
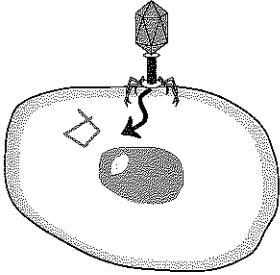
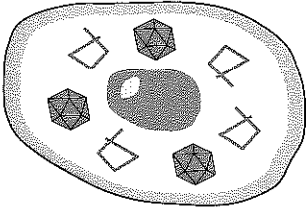
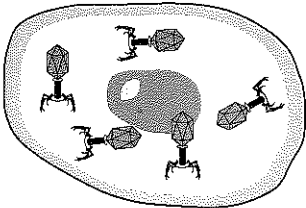
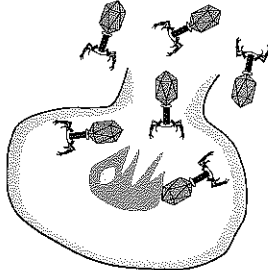
Check

- ☐ In pencil
- ☐ Ruled label lines
- ☐ Labels
- ☐ Continuous lines
- ☐ No shading

continued ...

Mix and match (optional activity)

Cut out the cards, mix them up and then match them.

Stage	Description	Diagram
Attach	The virus attaches to the host cell surface.	
Inject	The virus injects nucleic acid (DNA or RNA) into the host cell.	
Replicate and synthesise	The virus nucleic acid replicates and makes protein for capsules.	
Assemble	The virus nucleic acid assembles new viruses.	
Release	The virus nucleic acid releases viruses.	

Vaccines

Learning objective: To identify what a vaccine contains.

Key skills: Sort information.

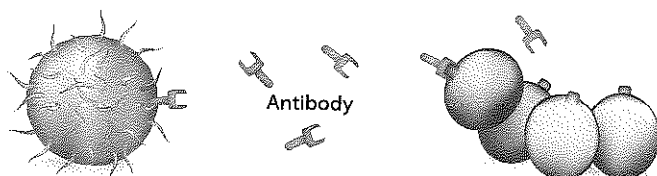
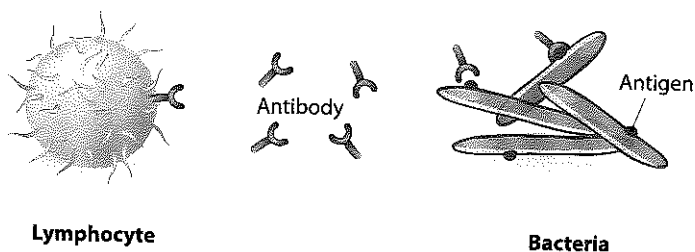
Materials: Scissors, glue stick, A4 paper.

Pathogens are micro-organisms such as viruses, bacteria, fungi, mould and yeasts that cause disease.

Vaccines can contain:

- live pathogens that have been made harmless
- harmless bits of the pathogen
- toxins made by pathogens
- dead pathogens.

Your immune system has white blood cells called lymphocytes. The lymphocytes make antibodies (proteins) that attach to antigens on the pathogen. The antibodies then destroy the pathogen.



Questions

1. Name five types of pathogen.

2. Name two things vaccines can contain.

3. Name two diseases that you can be vaccinated for.

Group activity

Complete this activity in a group of four.

1. You have 10 minutes to cut out the cards on the next page and then sort them into two groups: Trash (false) and Treasure (true).
2. Your group will have one minute to present one of the Trash cards to the class and explain how you would change it to make it true.

continued ...

Trash and treasure cards

Trash	You can't be immunised against a virus.
Treasure	Leukocytes or white blood cells take about two weeks to attack live or dead pathogens.
Leukocytes or white blood cells take one day to attack live or dead pathogens.	Different vaccines are needed for different pathogens.
Childhood vaccines give protection against many serious diseases.	You need vaccine boosters for some diseases because fewer white blood cells recognise the virus over time.
You can get sick because of the bits of inert virus in a vaccine.	When 90% of the population is immunised, unimmunised people will not catch the disease.
Vaccines work so well that many young people believe the pathogens cannot be deadly.	You do not need to be immunised for diseases such as tuberculosis, measles, mumps and rubella.
If you catch a live virus, it is better for your immune system than being vaccinated.	A very small number of people are allergic to the egg white used in vaccinations.
Vaccines are injected in egg white. People with albumen allergies should not be vaccinated.	Old or very young people or people with weak immune systems should not be vaccinated.

Polio

How?

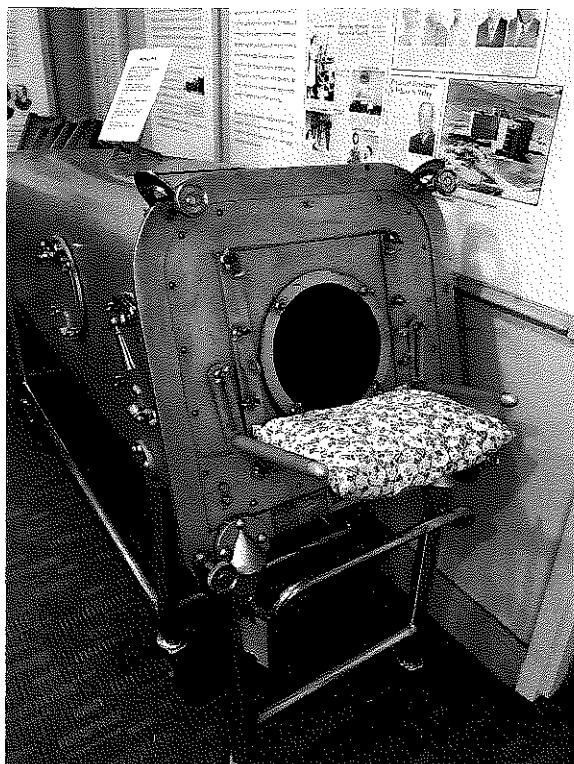
Learning objective: To identify the impact of vaccination.

Key skills: Interpret a table; draw a graph.

The full name for polio is **poliomyelitis**, a virus that causes paralysis (your muscles stop working). In the 1800s and early 1900s, many children had polio. Patients were kept in quarantine, away from other people. Patients with paralysed lungs lay inside an iron lung, which helped them breathe. Shelley Fraser Mickle, who got polio as a child, tells of her memories while in hospital:

Whole rooms of iron lungs housed children in one ward, adults in another. Only their faces showed, reflected in mirrors overhead. The sound was as if vacuum cleaners were hooked to the people's lungs. A terrific suck in, as the air moved, and then a hard push out as the machines breathed for those who couldn't. (For more of Shelley's story, see <https://www.npr.org/templates/story/story.php?storyId=5337953>)

In 1921, at the age of 39, the future American president Franklin D Roosevelt was paralysed by polio and had to use a wheelchair for the rest of his life.



Iron lung



Statue of US President Franklin Roosevelt, Washington DC

Photos: Left © Sgerbic; licensed for reuse under CC BY-SA 4.0 Right © jimbowen0306; licensed for reuse under CC BY 2.0

In the United States of America in 1953, Dr Jonas Salk discovered a polio vaccine. In 1954 and 1955, he carried out a **fair test**. About 2 million American school children were injected with either a polio vaccine or a **placebo** (a control containing no vaccine). No vaccinated child caught polio. Today, no one has polio in countries where nearly everyone (more than 80%) is vaccinated. The following table shows how the number of people with polio fell as the percentage who were vaccinated against it increased in the USA.

continued ...

Number of people per thousand with polio and percentage getting polio vaccination in the USA 1942–1965

Year	People with polio per thousand	% people vaccinated	Year	People with polio per thousand	% people vaccinated
1942	120	0	1954	150	70
1943	190	0	1955	80	80
1944	140	0	1956	30	90
1945	260	0	1957	40	80
1946	110	0	1958	50	80
1947	280	0	1959	30	90
1948	420	0	1960	8	90
1949	340	0	1961	1	90
1950	290	0	1962	0.4	90
1951	580	0	1963	0.1	90
1952	360	0	1964	0.05	90
1953	190	0	1965	0.01	90

- On the next page, draw a graph showing the data from the table. On the y axis, plot 0, 100, 200, 300, 400, 500, 600. Plot the number of people with polio per thousand as a bar graph, and the percentage of people vaccinated as a line.
- In what year did the most people have polio? How many people had it?

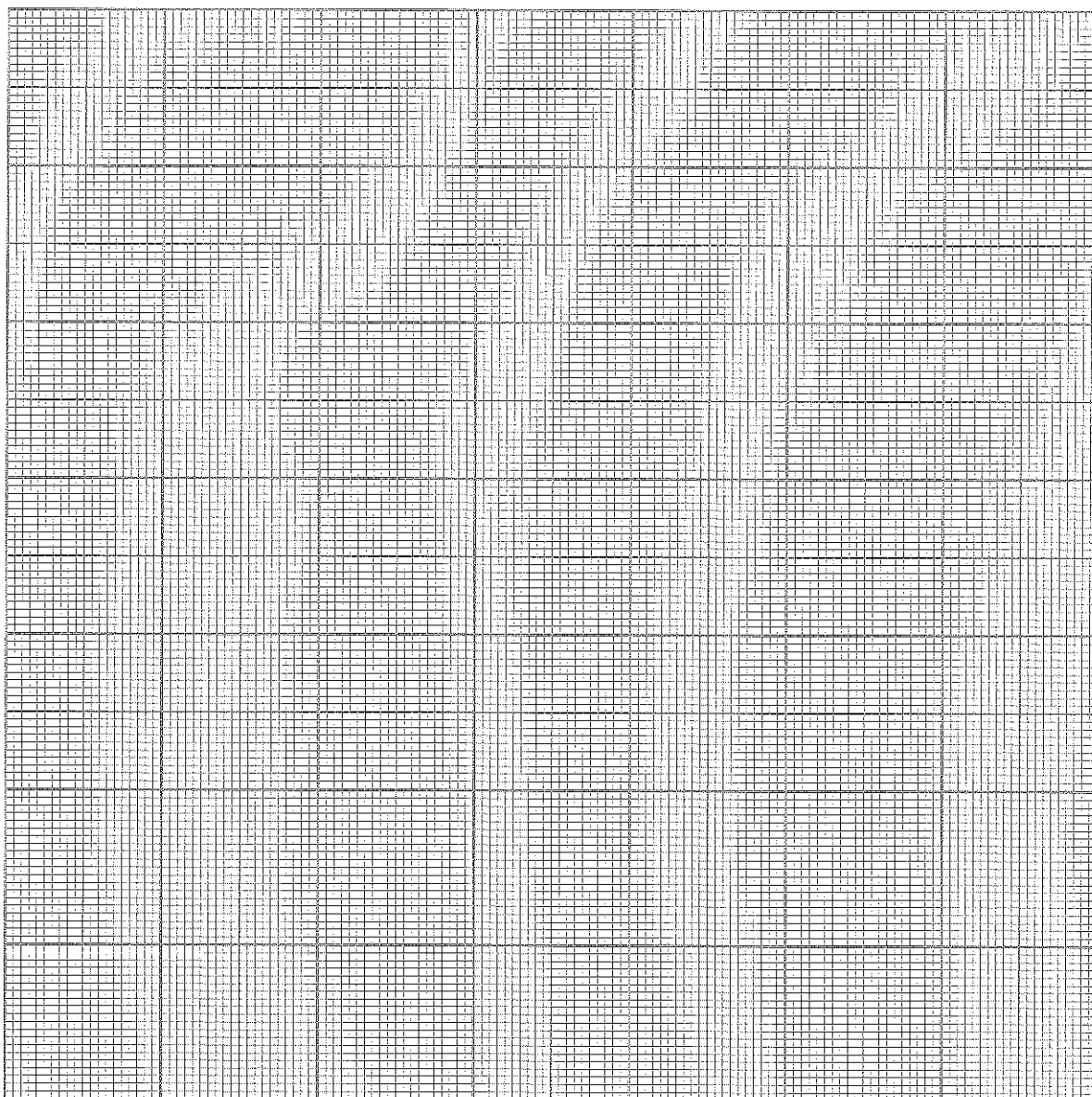
- In what year did polio vaccinations begin in the USA? How many people had polio in that year?

- How many people had polio 10 years after vaccination had begun?

- What trend do you notice once 90% of Americans started getting vaccinated against polio?

- Why is it important to be vaccinated?

continued ...

**Check**

- ☐ The graph is in pencil.
- ☐ The x and y axes are ruled.
- ☐ The graph is at least half a page.
- ☐ The graph has a title with the independent and dependent variables and units.
- ☐ The scale on the x axis is in even intervals (eg, 5, 10, 15 grid squares) to match the variable intervals.
- ☐ The scale on the y axis is in even intervals.
- ☐ The x axis is labelled with the variable and units.
- ☐ The y axis is labelled with the variable and units.
- ☐ A best-fit line is ruled through all points, although it may not touch all of them.

Bacteria

Learning objective: To explore the characteristics of bacteria.

Key skill: Conduct an experiment.

Materials: See experiment instructions.

Testing for bacteria

Let's talk ... to the doctor

You: Doctor, I'm not feeling well.

Doctor: That's no good. Open up and say "aaah!" (*Looks in your throat.*) I'll just take your temperature. (*Places thermometer in your ear.*) Now I'll listen to your chest. (*Places stethoscope on back.*) Breathe in ... out. Hmm. Would you mind coughing up some **mucus** into this sterilised container? I need about a teaspoon.

You: Crrgh! Spit!

Doctor: (*Screws on the lid and writes your name and date on the container.*) Thanks. I'll send it to the laboratory for testing.

You: What are you testing for?

Doctor: There are lots of white blood cells in your mucus (phlegm, sputum), which makes it look yellow. The white blood cells are fighting an infection from a **pathogen** (disease-causing micro-organism). I want to find out if the pathogen is a virus, bacterium, fungus or something else. Antibiotics kill most pathogens except viruses.

You: What do the scientists do at the lab?

Doctor: At the lab, scientists wear face masks and surgical gloves to prepare a **Petri dish**. If they breathe on or touch the dish with bare hands or leave the lid off for more than two minutes, they could add pathogens to it that you don't have. They pour in a jelly called **agar**, which contains **nutrients**. When the agar sets, the scientists quickly **inoculate** it using a cooled, **sterilised** wire dipped into your mucus. A hot wire would kill the pathogen. The scientists cover the dish with a lid, seal the lid with sellotape, and put it into a warm incubator upside down to stop condensation from dripping on the agar. The pathogens feed on the nutrients and grow. For up to eight weeks, the scientists check the dishes every day to see what bacterial or fungal **cultures** are growing. They always keep the lids on and throw away the dishes as there is a small possibility that you may have a very dangerous pathogen.

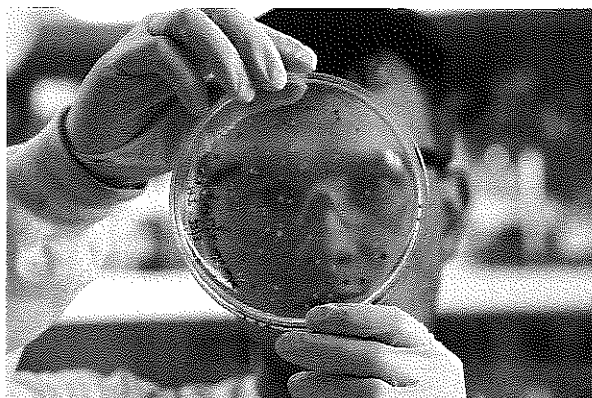


Photo: © CSIRO; licensed for reuse under CC BY 3.0

continued ...

BACTERIA

1. List at least three things that the doctor and scientists do to make sure that only your pathogens are cultured.

2. Why is your sputum coloured and not clear?

3. What does the bacteria feed on?

4. Why is the Petri dish inverted (turned upside down)?

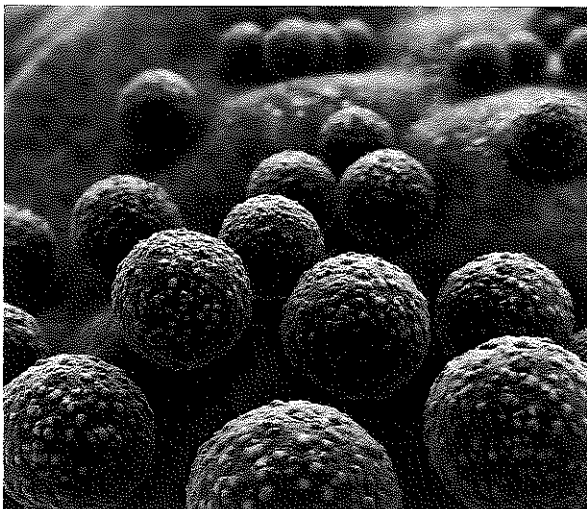
5. What grows on the agar?

Bacteria in summary

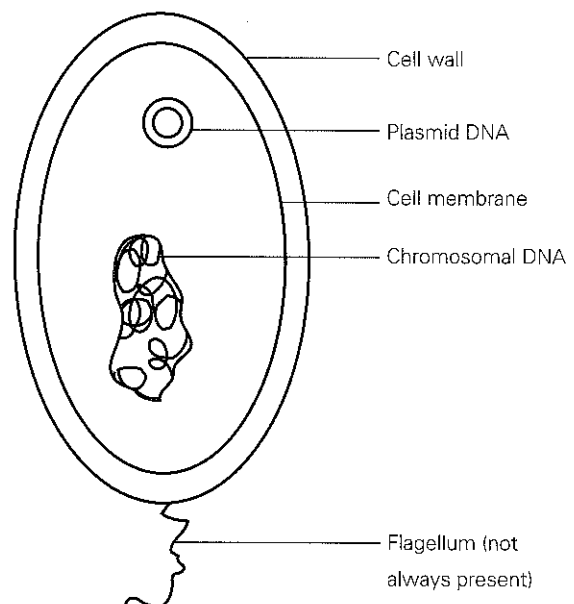
A bacterium:

- is a prokaryote
- consists of one cell with a cell wall and cell membrane
- has no nuclear membrane
- is very small at $0.2\text{--}10$ microns (0.0000002 metres)
- contains a chromosome with DNA
- contains circular plasmids (small DNA molecules)
- may have a flagellum (a tail that helps it move) or lots of flagella.

Note: We use *bacterium* to mean just one of this kind of microbe and *bacteria* for two or more.



Staphylococcus aureus bacterium under a microscope.
Photo: © www.scientificanimations.com; licensed for reuse under CC BY-SA 4.0



Experiment

Aim: To compare bacteria on hands that are washed, not washed and disinfected.

Alternative aim (optional): To compare bacteria in different parts of the school, for example, by a rubbish bin, in the toilets and on a desk.

Hypothesis

There will more bacteria on _____ than _____ and the fewest bacteria on _____

Materials

Petri dishes containing sterilised nutrient agar, disinfectant, soap, marker pen, inoculating wire (or cotton bud), incubator, Bunsen burner, matches, ziplock bags.

Note: Store Petri dishes upside down and bring to room temperature before inoculating.

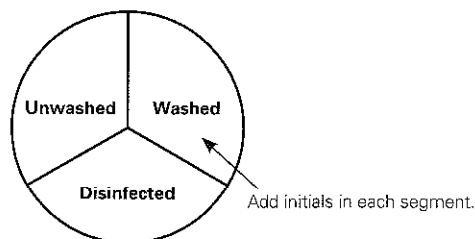
Safety first: Complete this sentence by circling the word in bold of your choice.

The hole of the Bunsen burner should be **open / closed** before I light it.

Method

1. (Circle the words in bold of your choice.) Keep one Petri dish sellotaped closed. This is the **control / experiment** that makes the test **fair / unfair**.

2. In pairs, on the base of the Petri dish, use a marker pen to make it look like the picture on the right. In each segment, also include the initials of who (you or your partner) is going to have their hand swabbed for that condition.



Option: Label the parts of the school that you collect bacteria from, eg, *Rubbish bin, Toilet, Desk*.

3. Sterilise your inoculating wire over a blue flame until red hot. Cool.

(a) Why do you need to sterilise the inoculating wire?

(b) Why must you not touch or breathe on the inoculating wire?

(c) Why do you need to cool the inoculating wire?

4. Wash your hands for 20 seconds in soapy water and then dry them for 20 seconds. Use the inoculating wire to swab your partner's unwashed hand.

Option: Swab different parts of the school, eg, rubbish bin, toilet, desk.

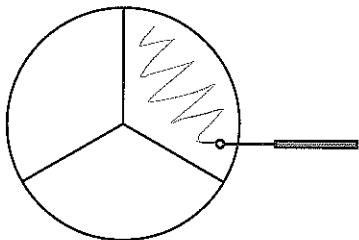
5. Take the lid off the Petri dish.

(a) Why should you *not* breathe on the agar or touch it with bare hands?

continued ...

(b) Why should the Petri dish lid be off for less than two minutes?

6. Touch the inoculating wire gently on the agar and make a zigzag pattern in the way shown in the picture below. Do not dig the wire into the surface. Put the lid back on.



7. Wash your hands and dry well with a paper towel. Now your partner swabs your hand and repeats step 5.
8. Your partner disinfects their hands. You swab your partner's hand and repeat step 6.
9. Sellotape the lid closed. Turn the dish upside down and put it in an incubator for three days.

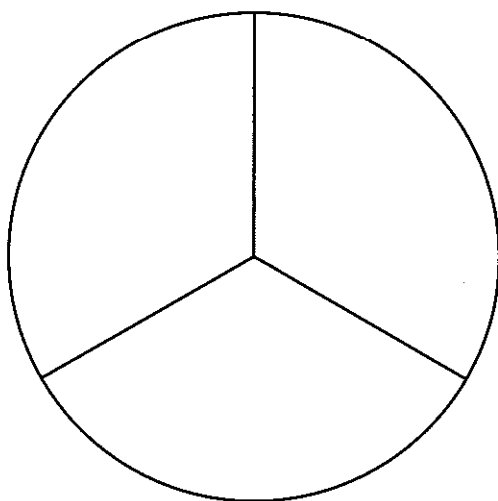
10. **Safety first**

(a) Once bacteria are in culture, why do you *not* open the Petri dish?

(b) After you have your results, dispose of the Petri dish in a ziplock bag.

Results

Draw your results.



Check

- ☐ Sharp pencil
- ☐ Continuous lines
- ☐ No shading
- ☐ Title
- ☐ Labels
- ☐ Ruled lines to labels
- ☐ Labels on right side of page

Conclusion

1. More bacteria were on _____ than on _____ and the fewest bacteria were on _____
2. Why did you need to store the Petri dish upside down?
-
-

Pathogens

Why?

Learning objective: To investigate some viruses and bacteria and why they make us sick.

Key skill: Summarise information.

Most micro-organisms are not harmful. A **pathogen** is a virus, bacterium, fungus, protozoa, parasite or other micro-organism that can cause disease. In everyday language we call pathogens *germs* or *bugs*. All pathogens can infect a host and avoid the immune system.

Your teacher will put resource cards on common pathogens caused by viruses and bacteria around the classroom. The pathogens are:

- measles
- chicken pox
- mumps
- rubella (also called German measles)
- tetanus (also called lockjaw)
- influenza
- hepatitis
- rabies
- *E coli* and campylobacter
- athlete's foot.

Choose one of these options as the subject for your activity:

1. symptoms (in a group of three or four)
2. spread and prevention (in a pair or on your own).

Option 1: Symptoms

In a group of three or four, choose one of the pathogen resource cards. Study the facts about it and prepare a two-minute play in which you:

- act out the symptoms
- say whether the disease is a virus, fungus or bacterium.

Then present your play to the class. The audience must guess which pathogen it is about.



Image: Released into the public domain under CC0 1.0

continued ...

Option 2: Spread and prevention

On your own or in a pair, use the resource cards to complete the following table. It has been started for you.

Type of pathogen	Tick one			Spread by	Prevented by
	Virus	Bacterium	Fungus		
Measles					Vaccine Quarantine
Chicken pox				Coughing Sneezing	
Mumps	✓				
Rubella					
Tetanus					
Influenza					
Hepatitis					
Rabies					
<i>E coli</i> and campylobacter					
Athlete's foot					

continued ...

Measles

Do you have a fever, runny nose, sore red eyes and white spots inside the mouth? After a few days, has a red blotchy rash appeared and then it lasts for up to one week? The rash usually starts on the face and spreads to the rest of the body.

Measles is the **rubeola virus** that spreads easily from person to person through the air, such as when you breathe, cough or sneeze, or touch a person or object. Mothers who have measles can infect their unborn babies.

Measles can cause miscarriage in pregnant women, make you deaf or give you seizures and brain damage. It can even kill you.

The incubation period is one to two weeks. The infectious period is from five days before until five days after a rash appears.

In 2017, New Zealand had no measles or rubella because of vaccinations. However, visitors still bring the pathogen into the country. In 2018, fewer than 70% of young people aged 10 to 29 years were immunised against measles and some of those who were not immunised caught measles. In Australia, more than 80% of people are immunised and very few people catch measles.

Anyone with measles needs to be isolated – that is, they have to stay at home and miss out on things like school, work, sporting competitions and social events. Catching measles or having a vaccination gives you immunity for more than 20 years.



Child with measles. Photo: Released into the public domain under CC0 1.0

continued ...

Chicken pox

Do you have a fever, headache, runny nose and cough? Do you feel tired but not hungry? After one or two days, does a red rash appear and then it continues to spread for three or four days? Does the rash begin on the face and scalp, spread to the chest, back and stomach and then move to the arms and legs?

Is the rash itchy? It is important not to scratch the spots as they can become infected with bacteria.

Do blisters form with a yellow liquid that turns cloudy? The blisters pop to release the virus and then form scabs that fall off after one to two weeks.

Chicken pox is the **varicella** virus that spreads easily from person to person through the air when you breathe, cough or sneeze, or touch a person or object. Mothers with chicken pox can infect their unborn babies.

The incubation period is two to three weeks. The infectious period lasts from one or two days before the rash to one day after the blisters have dried (which takes five to seven days). If you have more than 50 spots, the rash can be inside the ears, nose, mouth, eyelids and eyes.

Anyone with chicken pox needs to be isolated – that is, they have to stay at home and miss out on things like school, work, sporting competitions and social events. Catching chicken pox or having a vaccination gives you immunity for about 50 years. Once you catch the virus, it lives in your nerves, so in later life, as your immune system weakens, you may get shingles.



Boy with chicken pox. Photo: © Phyllis Buchanan; licensed for reuse under CC BY-SA 2.0

continued ...

Mumps

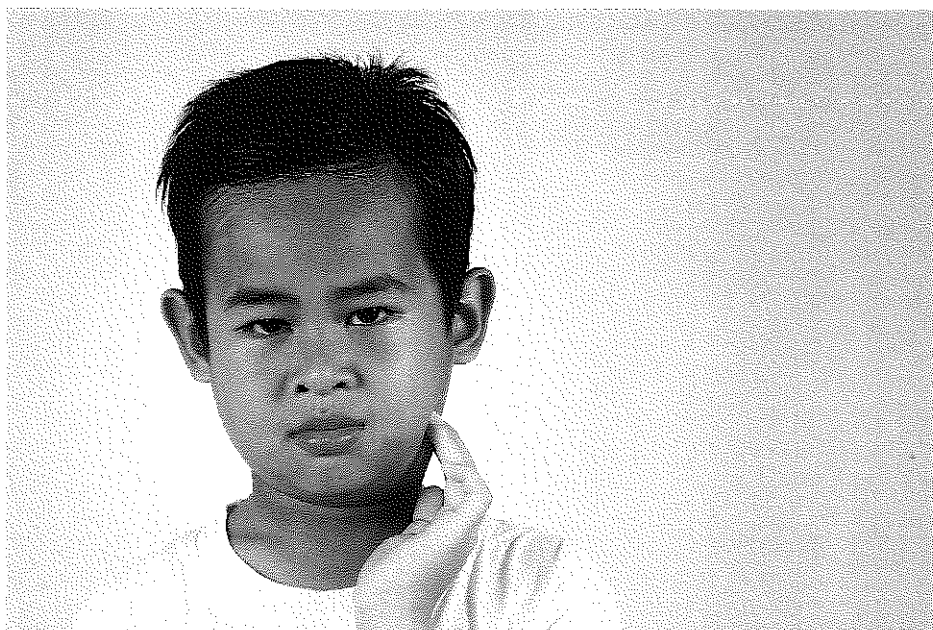
Do you have a slight fever, trouble breathing and swollen salivary glands (the parotid glands) below your ear? Even if you do not have any of these symptoms, it is still possible that you have the mumps.

Mumps is the **rubulavirus** that spreads easily from person to person through the air when you breathe, cough or sneeze, or touch a person or object. Mothers who have mumps can infect their unborn babies.

If you find you have mumps, you most likely came into contact with someone who had the virus about two to four weeks ago. You are infectious from a few days before you started feeling unwell until about five days after your glands start to swell. You stop being infectious about three to ten days after that.

Anyone with mumps needs to be isolated – that is, they have to stay at home and miss out on things like school, work, sporting competitions and social events. Mumps can lead to meningitis, deafness and (for males) swelling of the testicles. Catching mumps or having a vaccination gives you immunity for about 50 years. We can see how well vaccination helps to prevent mumps from United States data:

- In 1964, before vaccination began, the country had 212,000 cases of mumps.
- Since mumps vaccination began in 1967, cases of mumps have dropped by 99%.¹
- The quality of the vaccine affects the number of cases of mumps. For example, in 2009, when the quality of the vaccine was poor, the USA had 3,000 cases whereas in 2012, when the quality of the vaccine was better, 229 cases occurred.²



Boy with mumps

- 1 College of Physicians of Philadelphia (2018). Mumps. The History of Vaccines. URL: <https://www.historyofvaccines.org/content/articles/mumps>
- 2 AE Barskey, JW Glasser and CW LeBaron (2009) Mumps resurgences in the United States: A historical perspective on unexpected elements. *Vaccine* 27(44): 6186–95. URL: <https://www.ncbi.nlm.nih.gov/pubmed/19815120>

continued ...

Rubella (German measles)

Do you have a three-day rash over a lot of your body? Do your neck and armpits feel tender? Perhaps you have a mild headache, runny nose, reddish eyes and fever? Even if you do not have any of these symptoms, it is still possible that you have rubella.

The virus **rubella** spreads easily from person to person through the air when you breathe, cough or sneeze, or touch a person or object. Mothers who have rubella can infect their unborn babies.

If you find you have rubella, you most likely came into contact with someone who had the virus about two to three weeks ago. You will be infectious from one week before the rash appears or you feel unwell until about one week after the rash appears.

Rubella is a mild disease in children and adults but can cause rubella syndrome in unborn babies that have been in the womb for less than 20 weeks. Rubella syndrome can cause stillbirth or infected babies can be born with defects, for example, deafness, eye conditions such as cataracts, and heart defects.

Anyone with rubella needs to be isolated – that is, they have to stay at home and miss out on things like school, work, sporting competitions and social events. Catching rubella or having a vaccination gives you immunity for about 50 years.



Young woman with rubella

continued ...

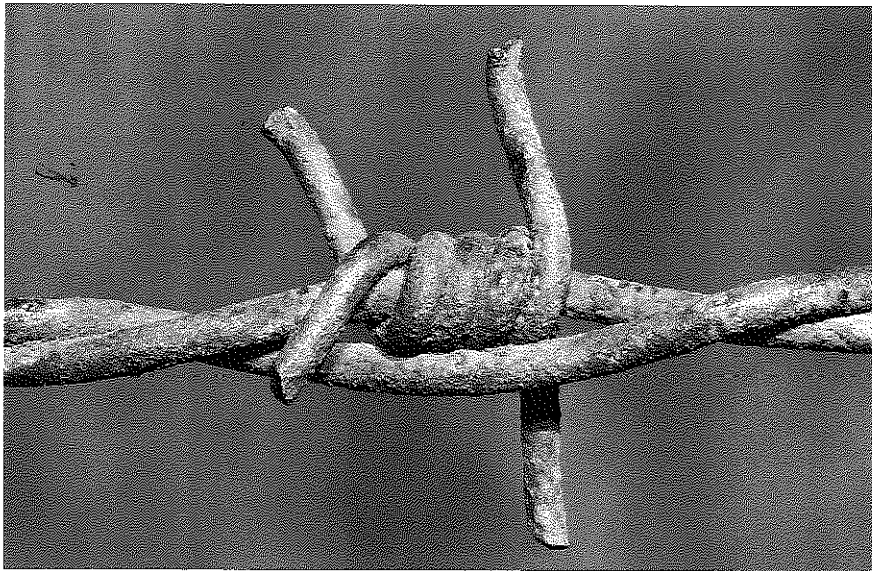
Tetanus (lockjaw)

Do you feel weak and stiff? Do you have cramps and find it hard to chew and swallow?

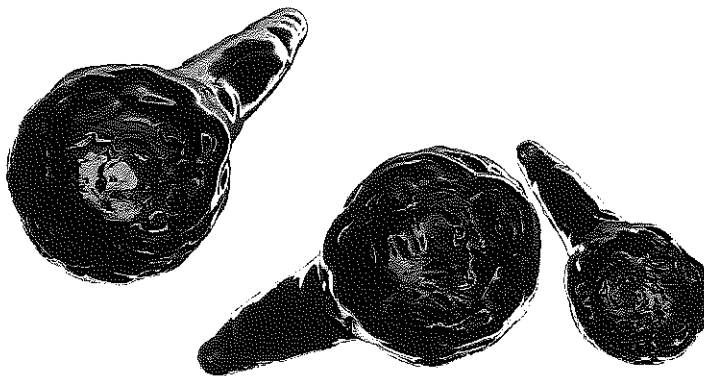
Tetanus is caused by the bacterium ***Clostridium tetani***, which is carried in the intestines of animals, including humans. *Clostridium tetani* is everywhere, but common in soil, and enters the body through wounds. Some common ways of catching tetanus include running barefoot on a farm and stepping on a nail or piece of wire, and cutting yourself on something rusty while gardening.

The tetanus bacterium makes a toxin that stops nerves from working properly. People with the tetanus toxin cannot relax their muscles. If untreated, the lungs become paralysed and the person dies.

Anyone with tetanus needs to be isolated – that is, you have to stay at home and miss out on things like school, work, sporting competitions and social events. When you have a tetanus vaccination, your immunity does not last for your whole life. You will also need booster tetanus vaccinations.



Wear shoes and other clothing to protect you on a farm and in other places where you might come into contact with rusty wire or nails. Photo: © Waugsberg; licensed for reuse under CC BY 2.0



The *Clostridium tetani* bacterium, which causes tetanus

continued ...

Influenza (flu)

Do you have a fever, headache, runny nose and cough?

The flu is the **influenza** virus that infects your nose, throat and lungs. It spreads easily from person to person through the air, when you breathe, cough or sneeze, or when you touch a person or object.



Woman with influenza. Photo: Released into the public domain under CC0 1.0

Very occasionally, the flu can cause serious illness and death.

If you find you have the flu, you most likely came into contact with someone who had the virus about one to four days ago. You will be infectious from one day before symptoms appear until up to seven days after becoming sick.

A number of different strains of the influenza virus can cause flu. Every year the vaccine is changed to include the most common strains of the virus.

Anyone with flu needs to be isolated – that is, they need to stay at home and miss out on things like school, work, sporting competitions and social events. If you have a flu vaccination, you may have immunity for a few virus strains for up to 90 years but for many strains you need to get a vaccine every year.

•

continued ...

Hepatitis

Do you feel exhausted, aching, and sick to your stomach but not hungry? Have you got a fever and a pain on the right side of your stomach near your ribs? Is your skin or are the whites of your eyes (sclera) slightly yellow (jaundiced)?

The yellow colour occurs because the liver has partly (but not completely) broken down your red blood cells.

The hepatitis virus **orthohepadnavirus** has different strains – A, B, C and D – which attack the liver.

If you find you have hepatitis, you probably shared body fluids, water or food with someone who had the virus. This may have involved sharing toothbrushes, having body piercings or tattoos, drinking water contaminated by faeces or urine, or handling blood. Mothers can pass the virus to their unborn babies.

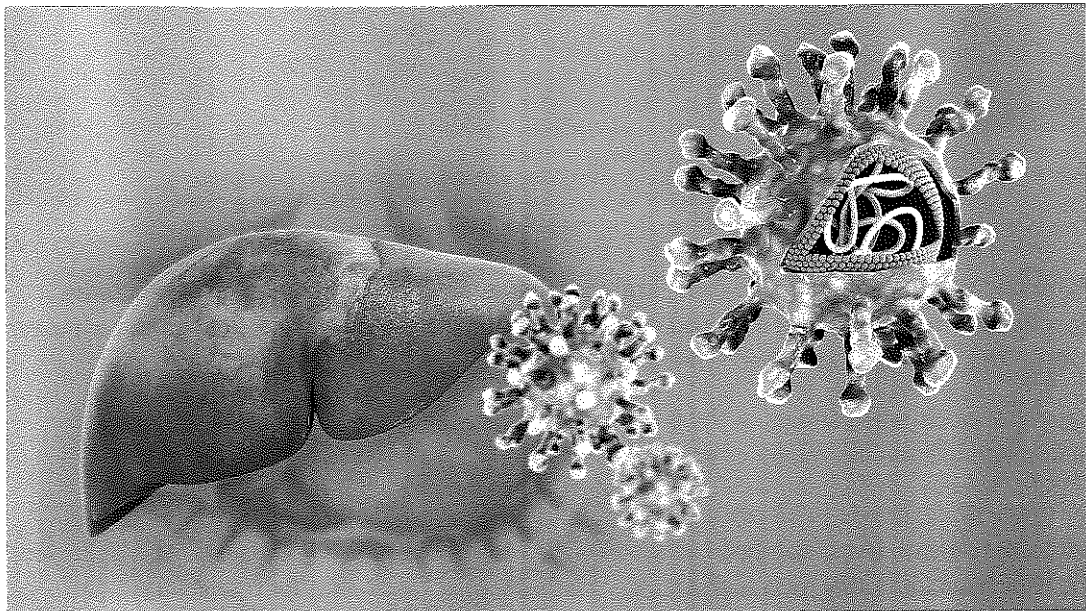
How long ago you got infected depends on the strain of the virus. It would have been:

- about two to six weeks ago for the A virus
- about six to twenty-three weeks ago for the B virus
- about two weeks to six months ago for the C virus.

How long you stay infected also depends on the strain. You will be infectious:

- for three weeks with the A virus
- for several weeks to six months with the B virus
- for six months or more with the C virus.

If you have hepatitis or a hepatitis vaccination, you have immunity for up to 90 years.



Hepatitis C virus. Image: © Manu5; licensed for reuse under CC BY-SA 4.0

continued ...

Rabies

Have you been bitten by an animal? Do you feel a tingling, prickling or burning sensation in the wound? Do you have a fever or headache? Are you vomiting, feeling nauseous or having difficulty swallowing? Are you foaming at the mouth, confused, frightened of water, over-excited or partly paralysed?

Rabies is the virus **lyssavirus**, which is spread through saliva. It infects the brain and then spreads through the nerves. If you have been licked or scratched by an animal that has rabies, wash any wounds for at least 15 minutes in soapy water, detergent or iodine, any of which can kill the virus. Rabies from a bite or deep wound will kill the infected person.

The incubation period is usually one to three months, but can be anywhere from one week to a year. The infection period is 10 days; after that, the infected person (or other animal) dies.

Dogs are most likely to spread rabies, but other animals – such as foxes, deer, bats and skunks – can too. Any mammal can catch rabies, including humans, cats, cows, goats and horses. Vaccination of dogs has reduced human rabies deaths in Sri Lanka from more than 350 in 1973 to 50 in 2010, and in Thailand from 170 in 1991 to 7 in 2011.



Keep away from stray dogs, which are more likely to have rabies. Photo: Released into the public domain under CC0 1.0

continued ...

E coli and *Campylobacter*

Have you got stomach cramps and diarrhoea? Have you got a fever, along with nausea and vomiting? You may have food poisoning from *E coli*, *Campylobacter*, *Yersinia*, *Shigella* or other bacterial infections of the intestine.

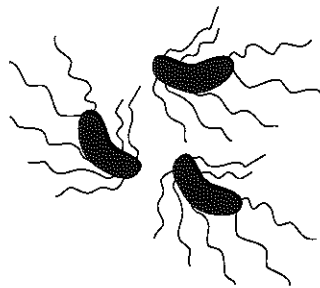
Escherichia coli

Escherichia coli (*E coli*) bacteria live in the intestine. They are passed on in raw chicken and by touching faeces (poos) or by drinking water that contains faeces. If you find you have *E coli* infection, about three to four days ago (although it can be anything up to 10 days ago) you may have drunk water from a stream that runs through a farm or touched faeces and then did not wash your hands properly before you ate. Most people recover within a week to 10 days. You stop being infectious one to two days after you feel well again.

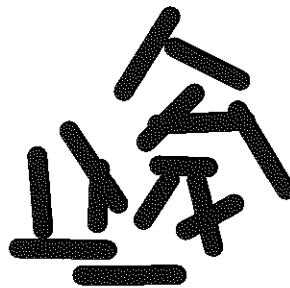
Campylobacter

Campylobacter bacteria live in the intestine. They are passed on in unpasteurised milk or raw chicken and by touching faeces or by drinking water that contains faeces. The *Campylobacter* infection is spread when people use the same knife and chopping board for meat and vegetables, or do not wash their hands after handling raw meat.

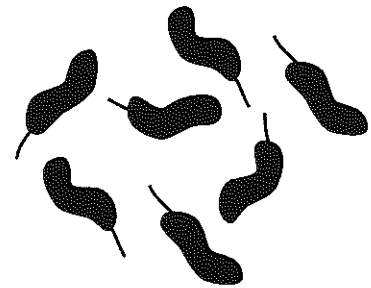
If you find you have a *Campylobacter* infection, you may have eaten bad food in the past two to five days. Most people recover within two to ten days. You stop being infectious one to two days after you feel well again.



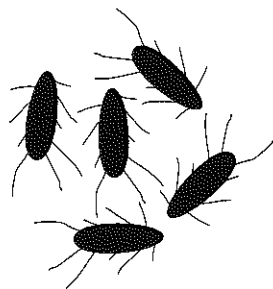
Salmonella



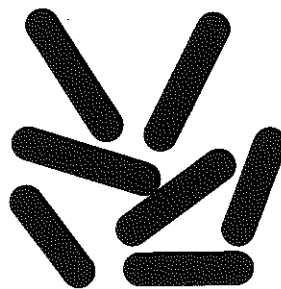
Yersinia



Campylobacterium



E coli



Shigella

Bacterial infections that affect the intestine

continued ...

Athlete's foot

Have you got peeling skin between your toes or on the soles of your feet? Are your toenails white or coloured? Do your feet itch?

You may have a skin infection caused by a fungus yeast commonly called **athlete's foot**. Many different species of fungi cause athlete's foot.

Fungi grow best in warm, dark, damp areas and need oxygen to respire.

The fungus is contagious and spreads from surfaces such as damp floors, towels, clothing and shoes. The incubation period is unknown, and the period of infection can go on forever, depending on what treatment you use and whether you practise good foot hygiene.

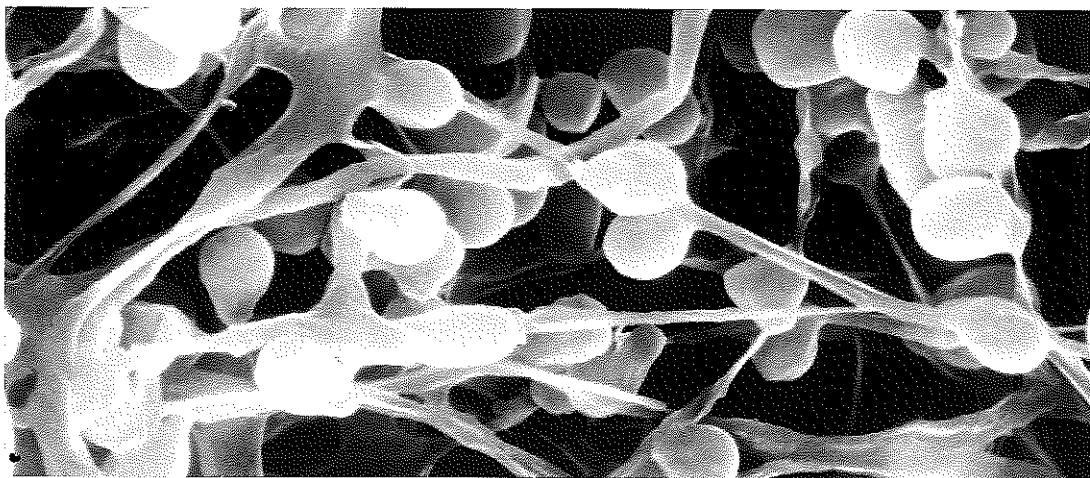
To avoid infection, wear jandals at public swimming pools and showers, dry between your toes, keep your toenails cut level, take your shoes and socks off as soon as you can and change your shoes and socks daily.

If you have a nail infection, use a fungicide toenail polish. Remove and reapply this every other day. The nail polish is transparent (lets in light) and removes air from the surface of the nail.

If you have a skin infection, use a fungicide powder or cream daily for four weeks as the fungus can live in the skin even when the symptoms have gone.



Athlete's foot. Photo: © ProjectManhattan; licensed for reuse under CC BY-SA 3.0



Athlete's foot fungus. Photo: © Ecorahul; licensed for reuse under CC BY-SA 4.0

Ginger beer



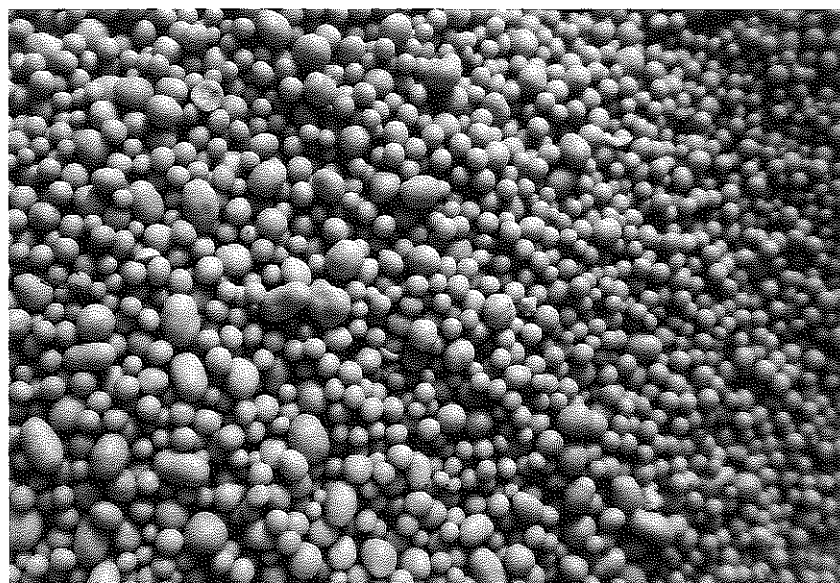
Learning objective: To investigate the characteristics of a fungus.

Key skill: Conduct an experiment.

Materials: See experiment instructions.

Features of a fungus – yeast

- Eukaryote
- One cell
- Cell nucleus
- Reproduces by fission, sexual reproduction or both



Brewer's yeast. Photo: Released into the public domain under CC0 1.0

Aim: To make ginger beer.

Materials

You will need to use a kitchen.

- 1 cup sugar
- 1/2 cup warm water
- 1/8 tsp active dry yeast or brewer's yeast
- Large pot
- 2 tbsp grated fresh ginger and/or lemon zest
- 3 tbsp lemon juice
- 7 C water, boiled then cooled

Method and results

Fill in the blanks to complete this method and results section, using the word bank to help you.

continued ...

Word bank

yeast	acidity	anaerobic	dissolved	energy
type	fermentation	boiled	acid	warm
respiration	kills	lower	firm	unwanted
respire	limits	time	carbon dioxide	

- Put (a) _____, sugar and water in a warm place to start (b) _____. The reaction is (c) _____ and occurs because the yeast carries out (d) _____ (without oxygen) respiration. The yeast mixture foams as it begins to produce carbon dioxide.
- To release the (a) _____ from lemon peel, juice and ginger, steep (soak) these ingredients in boiling water. The lemon juice lowers the pH ((b) _____) and this stops many other microbes from growing. The boiling water also (c) _____ any unwanted bacteria or other microbes. Sugar is (d) _____ at this time.
- Add the foaming yeast mixture to the ginger mixture. The yeast feeds on the sugar and breaks it down for (a) _____ (respiration) releasing (b) _____ and alcohol. The amounts of each waste product vary depending on the (c) _____ of yeast used, the amount of sugar and the (d) _____ for fermentation. This recipe is non-alcoholic and will produce plenty of fizz (carbon dioxide).
- Add some (a) _____ then cooled water to make up the required volume. Again, you boiled the water first to kill any (b) _____ microbes.
- Leave the bottle of "ginger beer" in a (a) _____ place for a couple of days. Because it is warm, the yeast continues to survive and carry out (b) _____. The yeast in the bottle continues to feed on the sugar, (c) _____ and release waste products until all the sugar has been used or something else (d) _____ the yeast growth.
- When the bottle is (a) _____, it means that lots of carbon dioxide has been produced. At this stage, put the bottle in the fridge. Because of the (b) _____ temperature, the yeast fermentation slows down – no more carbon dioxide is produced. The ginger beer is now "fizzy" and ready to drink.

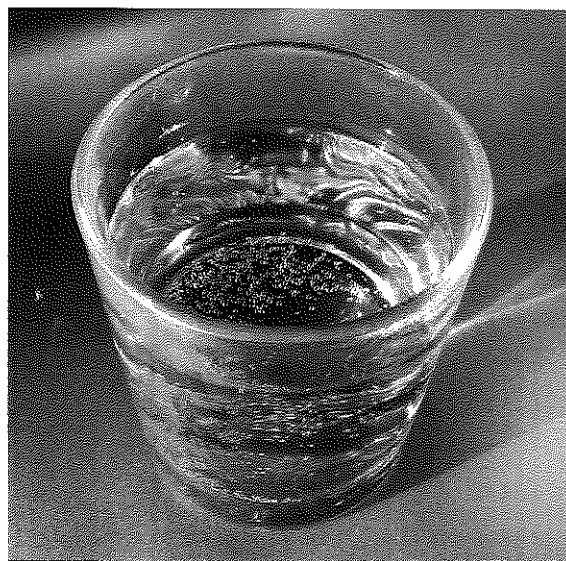


Photo: © GeoTrinity; licensed for reuse under CC BY-SA 3.0